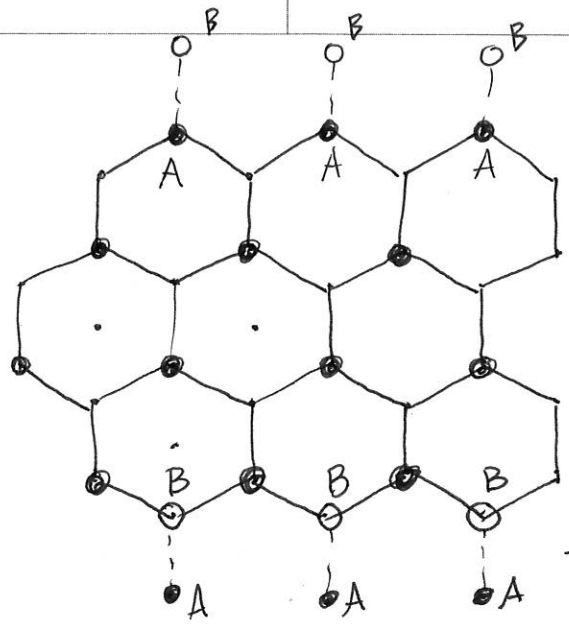
Zigzag edge



$y = \frac{L}{2}$ $\psi_B(\frac{L}{2}) = 0$

$y = -L/2$ $\psi_A(-\frac{L}{2}) = 0$

Semi-infinite graphene layer with zig-zag edge: boundary condition is that $\psi_B(x, y = \text{edge}) = 0$.

Recall that $\psi(x, y) = \begin{pmatrix} \psi_A(x,y) \\ \psi_B(x,y) \end{pmatrix}$.

$$H_K = \hbar v \begin{pmatrix} 0 & q_x - i q_y \\ q_x + i q_y & 0 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} \Rightarrow H \begin{pmatrix} u \\ v \end{pmatrix} = \varepsilon \begin{pmatrix} u \\ v \end{pmatrix}$$

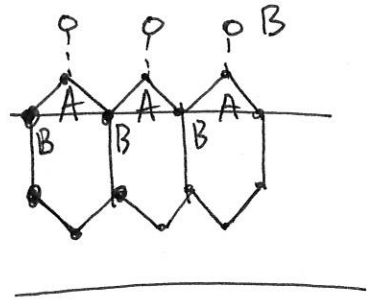
$$q_x = -i \frac{\partial}{\partial x}$$

$q_x - i q_y$
$$\left(-i \frac{\partial}{\partial x} - \frac{\partial}{\partial y}\right) v = \varepsilon u$$

$$\left(\frac{\partial}{\partial x} - i \frac{\partial}{\partial y}\right) v = i \varepsilon u \quad (1)$$

$q_x + i q_y$
$$\left(-i \frac{\partial}{\partial x} + \frac{\partial}{\partial y}\right) u = \varepsilon v$$

$$\left(\frac{\partial}{\partial x} + i \frac{\partial}{\partial y}\right) u = i \varepsilon v \quad (2)$$



Let $\begin{cases} u(x,y) = e^{i k_x x} u(y) \\ v(x,y) = e^{i k_x x} v(y) \end{cases}$

$$\Rightarrow \begin{cases} \left(i k_x - i \frac{\partial}{\partial y}\right) v(y) = i \varepsilon u(y) \\ \left(i k_x + i \frac{\partial}{\partial y}\right) u(y) = i \varepsilon v(y) \end{cases}$$

$$\begin{cases} \left(k_x + \frac{d}{dy}\right) u = \varepsilon v \\ \left(k_x - \frac{d}{dy}\right) v = \varepsilon u \end{cases}$$

Try $\begin{cases} u(y) = A e^{\lambda y} + B e^{-\lambda y} \\ v(y) = C e^{\lambda y} + D e^{-\lambda y} \end{cases}$ $\begin{cases} u(y = -\frac{L}{2}) = 0 \\ v(y = \frac{L}{2}) = 0 \end{cases}$ Boundary condition

$$\begin{cases} A e^{-\lambda L/2} + B e^{\lambda L/2} = 0 \\ C e^{\lambda L/2} + D e^{-\lambda L/2} = 0 \end{cases}$$

So BC $\psi_B(x, y=0) = 0 \Rightarrow v(x, y=0) = 0$

let $u(y) = e^{-\lambda y} u_0$, $v(y) = e^{-\lambda y} v_0$

but BC $v(x, y=0) = 0$ above means $\underline{v_0 = 0}$

Then $\left(K_x + \frac{d}{dy}\right) u(y) = (K_x - \lambda) u_0 e^{-\lambda y} = \varepsilon e^{-\lambda y} v_0 = 0$

$\Rightarrow K_x = \lambda$.

Then $\left(K_x - \frac{d}{dy}\right) \overset{=0}{v_0} e^{-\lambda y} = 0 = \varepsilon e^{-\lambda y} u_0 \Rightarrow \underline{\varepsilon = 0}$

Thus we have <u>zero-energy</u> solution ($\varepsilon = 0$)

$$u(x,y) = e^{iK_x x} e^{-K_x y} u_0$$

$$v(x,y) = e^{iK_x x} e^{-K_x y} v_0 = 0$$

(Normalizability requires $K_x > 0$ <u>only</u>)

which leaves on only one sublattice (A in this case)

$\Rightarrow$ zero-energy mode $\psi_K = \begin{pmatrix} e^{iK_x x} e^{-K_x y} u_0 \Theta(K_x) \\ 0 \end{pmatrix}$

where K_x is measured from $\vec{K}$. Observe that only positive $\underline{K_x > 0}$ is allowed $\longmapsto$ this mode only moves right!

Next, <u>At K'</u> : $H_{K'} = -\hbar v_F \begin{pmatrix} 0 & q_x + iq_y \\ q_x - iq_y & 0 \end{pmatrix}$

Solve $H_{K'} \begin{pmatrix} u \\ v \end{pmatrix} = \varepsilon \begin{pmatrix} u \\ v \end{pmatrix}$ subject to the boundary condition $\psi_B(x, y=0) = v(x, y=0) = 0$.

$$\begin{pmatrix} 0 & (-v_F)\left(-i\frac{\partial}{\partial x} + \frac{\partial}{\partial y}\right) \\ (-v_F)\left(-i\frac{\partial}{\partial x} - \frac{\partial}{\partial y}\right) & 0 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = \varepsilon \begin{pmatrix} u \\ v \end{pmatrix}$$

$$\begin{cases} v_F\left(i\frac{\partial}{\partial x} - \frac{\partial}{\partial y}\right) v = \varepsilon u \\ v_F\left(i\frac{\partial}{\partial x} + \frac{\partial}{\partial y}\right) u = \varepsilon v \end{cases}$$

Let $\begin{cases} u(x,y) = e^{ik_x x}\, u(y) \\ v(x,y) = e^{ik_x x}\, v(y) \end{cases}$

$$\begin{cases} \left(-k_x - \frac{d}{dy}\right) v(y) = \left(\frac{\varepsilon}{v_F}\right) u(y) \\ \left(-k_x + \frac{d}{dy}\right) u(y) = \left(\frac{\varepsilon}{v_F}\right) v(y) \end{cases}$$

$$\begin{cases} (1) \quad \left(k_x + \frac{d}{dy}\right) v = \tilde{\varepsilon}\, u \\ (2) \quad \left(k_x - \frac{d}{dy}\right) u = \tilde{\varepsilon}\, v \end{cases}$$

where $\tilde{\varepsilon} = \varepsilon / v_F$.

Again, $\begin{cases} u(y) = e^{-\lambda y} u_0 \\ v(y) = e^{-\lambda y} v_0 \end{cases} \Rightarrow$ due to BC $\underline{v_0 = 0}$

$$(2) \rightarrow \left(k_x - \frac{d}{dy}\right) e^{-\lambda y} u_0 = \tilde{\varepsilon}\, e^{-\lambda y} v_0 = 0$$

(2) $\left(k_x+\lambda\right) e^{-\lambda y} u_0 = 0 \Rightarrow \underline{\lambda = -k_x} \Rightarrow u(y) = e^{k_x y} u_0$

Then eq.(1) reads

$$\left(k_x + \frac{d}{dy}\right) e^{-\lambda y} v_0 = 0 = \tilde{\varepsilon}\, e^{-\lambda y} u_0$$

which is only possible if $\tilde{\varepsilon} = 0$.

Thus, again zero-energy solution

$$\psi_{k'} = \begin{pmatrix} e^{i k_x x} e^{k_x y} u_0 \\ 0 \end{pmatrix}$$

Here, since normalizability requires that $e^{k_x y} \xrightarrow[y\to\infty]{} 0 \Rightarrow$ only possible for negative $\underline{k_x < 0}$

$\Rightarrow \psi_{k'}$ describes left-propagating mode.

Full structure of $\varepsilon = 0$ mode due to BC $\psi_B = 0$

$$\psi = e^{i\vec{k}\cdot\vec{r}} \begin{pmatrix} e^{i k_x x - k_x y}\, \theta(k_x) \\ 0 \end{pmatrix} +$$

$$+ e^{i\vec{k}'\cdot\vec{r}} \begin{pmatrix} e^{i k_x x + k_x y}\, \theta(-k_x) \\ 0 \end{pmatrix}$$

$$\vec{k}' = -\vec{k}.$$